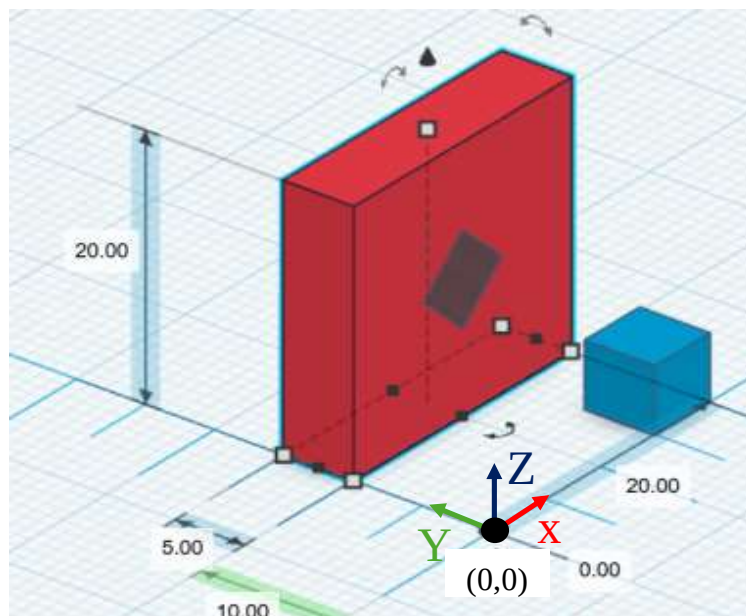


Introduction to Robotics

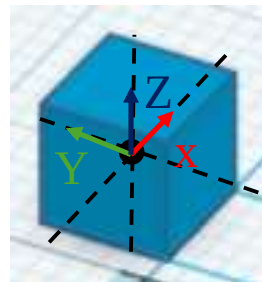
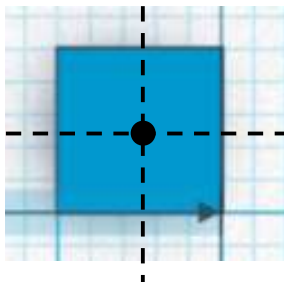
1st Homework

Tasks

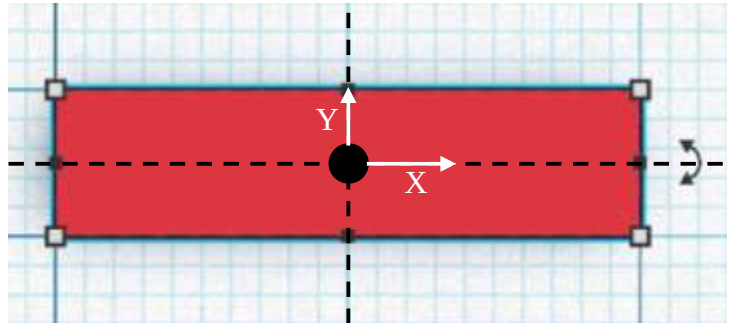
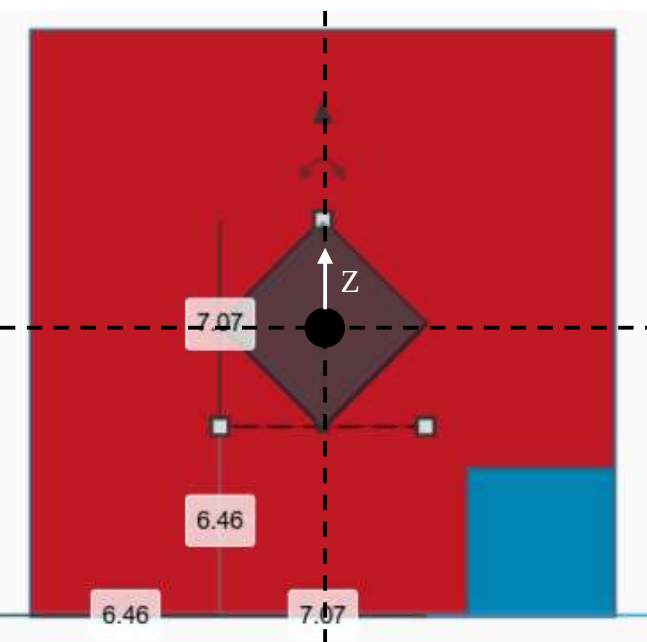
1. The marked point in the picture is the base of the world frame, the orientation goes as follows: x-axis pointing to the blue cube, y-axis pointing to the red object, z-axis pointing upwards.



2. The base point of the cube's body frame is the geometric center of the cube as shown below. The orientation of the body frame's axes is the same as the world frame.



2. The base point of the body frame of the red object is again at its geometric center as shown. The orientation of the axes is the same as the world frame.



3. To calculate the transformation matrix of each object relative to the world frame we need the coordinates of the bases of the body frames as seen from the world frame.

Cube:

We know that the bottom right outer corner of the cube is at $x = 20 \text{ cm}$ and that the cube has sides 5 cm long. Thus, its base point will be at $x = 20 - \frac{5}{2} = 17.5 \text{ cm}$. We can easily find that the point is at $P(x, y, z) = P(17.5, 2.5, 2.5)$. The transformation matrix consists of the Rotation Matrix, the vector p and the values 0 and 1 because we are talking about points and not vectors. The Rotation Matrix is equal to the identity matrix in this case because the world and body frame axes are aligned.

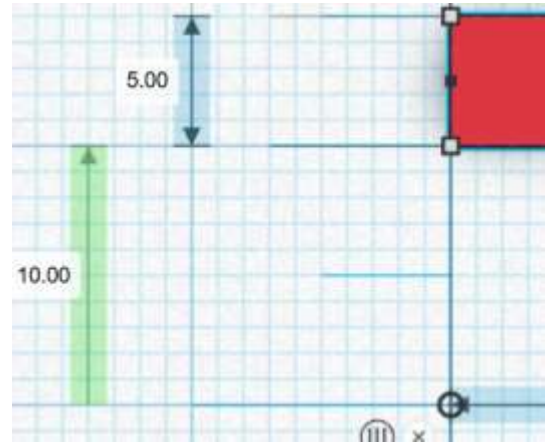
The vector p is $p = [17.5 \ 2.5 \ 2.5]^T$. So, the transformation matrix of the base frame of the cube relative to the world frame is:

$$T(w, \text{cube}) = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 17.5 \\ 0 & 1 & 0 & 2.5 \\ 0 & 0 & 1 & 2.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Red object:

We know that this object is 20 *cm* long (x-axis) so the center is at $x = 10$ *cm*. Its height is also 20 *cm* so its center is at $z = 10$ *cm*. Its width is 5 *cm* and so from the picture below we calculate the center's y ($y = 12.5$ *cm*). Like the transformation matrix of the cube, we have:

$$T(w, \text{box}) = \begin{bmatrix} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & 12.5 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



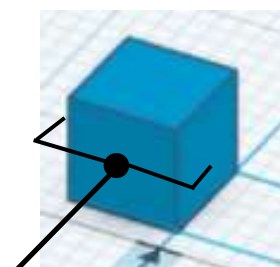
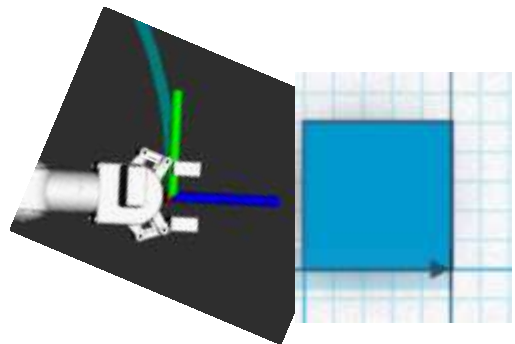
4. We mount the manipulator at $(x, y, z) = (0, 0, 0)$. Its base frame is, in that case, the same as the world frame so the transformation matrix is going to be a 4x4 identity matrix.

This position ensures that the manipulator can perform its task correctly, in other words the cube can be grabbed by the end-effector without a concern for the distance (max-distance > cube distance $\rightarrow 38 > 20$).

$$T(w, \text{manipulator}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5. Assuming the end-effector can successfully grab the cube horizontally, from its left and right side, we want to grab it from the closest side. The orientation would be like shown:

We see that the end-effector frame needs to be rotated about y-axis by 90 deg to obtain the cube frame. Doing the calculation in python, we get the final rotation matrix.



The base point of the cube's frame is at $(x, y, z) = (0, 0, 2.5)$ with respect to the end-effector's frame. Finally, the transformation matrix is:

$$T(\text{end-effector, cube}) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 2.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

6. To find the transformation matrix for the position right next to the hole we need the coordinates and the orientation of the cube's frame at that time.
- The x coord will be the distance from the y-axis to the center of the hole. The hole is at the center of the object so the distance will be $x = 20/2 = 10 \text{ cm}$.
 - The y coord will be the distance from the x-axis to the side of the red object minus half the cube's length, $y = 10 - 5/2 = 7.5 \text{ cm}$.
 - The z coord will be the distance of the cube's base point from the "ground", $z = 20/2 = 10 \text{ cm}$, 20 being the red object's height.

Assuming the cube is at this position when entering the hole, its axes will be like shown in the image.

We need to rotate the axes about the z-axis by 90 deg and then about x-axis by 45 deg to obtain the final orientation.

So, the transformation matrix is: (see python code)

$$T(\text{cube out, world}) = \begin{bmatrix} 0 & 1 & 0 & 10 \\ -\cos(\pi/4) & 0 & \cos(\pi/4) & 7.5 \\ \cos(\pi/4) & 0 & \cos(\pi/4) & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The transformation matrix when the cube is inside the hole is the same except the y-axis distance which is going to be the current distance plus a cube's length,

$$7.5 + 5 = 12.5 \text{ cm}$$

So, $T(\text{cube in, world}) =$

$$= \begin{bmatrix} 0 & 1 & 0 & 10 \\ -\cos(\pi/4) & 0 & \cos(\pi/4) & 12.5 \\ \cos(\pi/4) & 0 & \cos(\pi/4) & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

